08-07 第2篇

阅读理解（说明文） Why Does the Moon Change Shape?

① Tonight, if you look up, you may see a thin curved moon like a silver boat. Next week, the moon may look like half of a bright coin. Two weeks from now, it will be a full round ball, pouring pale light into every window. Then, slowly, it will shrink again, until one night it has seemingly disappeared. For thousands of years, humans have watched these changes and wondered: **why does the moon change its shape?** The real answer is surprising: **the moon itself does not change at all.** What changes is only the way we see it.

② First, we should remember three simple facts. One, the Earth moves around the sun. Two, the moon moves around the Earth — a full trip takes about 29.5 days. Three, the moon does not make its own light; it only **reflects** the light of the sun, the way a mirror reflects a lamp. Any side of the moon that is turned toward the sun is bright. Any side turned away is dark. The ball-shape of the moon itself never changes. What we call the "phases of the moon" are just different angles, from our place on Earth, at which we catch the lit side.

③ Let us walk through a full moon-cycle, step by step. When the moon is almost between the Earth and the sun, its dark side faces us. The moon is still fully lit — but the lit part is facing the sun, not us. We see nothing. This is the **new moon**. The sky above seems moonless, and ancient people sometimes believed the moon had been eaten.

④ A few nights after the new moon, the moon has moved a little in its journey around the Earth. From our angle, we start to see a thin slice of its sunlit side — this is the **waxing crescent**, the "silver boat" in the evening sky. Each night, the slice grows a little wider.

⑤ About a week after the new moon, the moon has travelled a quarter of its orbit. Now, from Earth, we see half of its lit side. This is called the **first quarter moon**, although to most of us it just looks like a half-moon.

⑥ The moon keeps moving. More than half of the sunlit side begins to face us, and we see a **waxing gibbous moon** — fatter than a half, but not yet fully round. At last, about 14 days after the new moon, the moon reaches the other side of the Earth, opposite to the sun. Now the entire lit half is pointing at us, and we see a **full moon**: a round, bright disc that can rise at sunset and stay in the sky all night.

⑦ After the full moon, the process runs in reverse. The lit face we can see becomes smaller each night. This "shrinking" phase is called **waning**. We move back through

waning gibbous, last quarter, waning crescent, and return at last to a new moon. The whole cycle takes about 29 and a half days — very close to one calendar month. The English word "month" even comes from the same root as "moon".

⑧ There is a second question people often ask: "Why do I see the same 'face' of the moon every night? Why do the dark patches never move?" This is because the moon **rotates** on its own axis at almost exactly the same speed as it orbits the Earth. One rotation = one orbit. As a result, the same side of the moon always points at us. The other side — often called "the far side" — was only photographed for the first time in 1959, by a Soviet spacecraft. It is not darker than the near side (both get equal sunlight), but from the Earth, we simply never see it.

⑨ Understanding moon phases helps in many practical ways. Farmers used to time their planting by the moon. Fishermen still check the full moon, because it is linked with stronger tides. Stargazers know that a full, bright moon makes it hard to see faint stars, so they choose "new moon" nights for careful sky-watching. Even writers and musicians borrow the moon's shapes to describe human feelings: fullness, emptiness, growth, loss.

⑩ So next time the moon surprises you — a thin crescent at breakfast, a huge orange disc at dinner — remember: the moon is always a whole ball of rock, nearly 400,000 kilometres away, quietly circling us. It does not grow or shrink; it does not hide, or come back. It simply turns with us in a slow patient dance between the Earth and the sun. **What we see as "change" is really only a change in our point of view.**

根据文章内容，选择最佳答案。

1. According to the passage, why does the moon seem to change its shape each night?

A. Because the moon really grows and shrinks every month.
B. Because the moon slowly eats itself and grows back.
C. Because we see different parts of the sunlit side from different angles during the moon's journey around the Earth.
D. Because the Earth is slowly cutting pieces off the moon.

2. When is a "new moon" seen in the sky?

A. When the moon is on the opposite side of the Earth from the sun, fully lit toward us.
B. When the moon is almost between the Earth and the sun; its lit side faces the sun, so we see nothing.
C. When the moon is cut in half by the Earth's shadow.
D. When there is heavy rain for several nights.

3. How long does one full cycle of moon phases (new moon → new moon) take?

A. About 7 days.
B. About 14 days.
C. About 29.5 days.
D. Exactly 365 days.

4. Why do we always see the same side of the moon from Earth?

A. Because the other side is covered in permanent black clouds.
B. Because the moon rotates on its axis in almost the same time as it takes to orbit the Earth.
C. Because the far side of the moon is broken off.
D. Because the far side has no sunlight at all.

5. Which sentence best expresses the main idea of the passage?

A. The moon is a magical living creature.
B. The moon is dangerous and should be avoided.
C. The moon is always a whole ball; what we see as "change" is really only a change in our point of view.
D. Only astronomers can understand moon phases.

答案与解析

1.
【答案】C
【题型】细节理解题
【定位】第②段
【解析】"phases of the moon are just different angles"。A/B/D 均错。故选 C。

2.
【答案】B
【题型】细节理解题
【定位】第③段
【解析】月在地日之间，亮面朝太阳。A 是满月；C/D 均错。故选 B。

3.
【答案】C
【题型】细节理解题
【定位】第⑦段
【解析】29.5 天。A/B 为半周期/部分；D 是地球绕日。故选 C。

4.
【答案】B
【题型】细节理解题

【定位】第⑧段
【解析】自转周期≈公转周期。A/C/D 均错。故选 B。

5.
【答案】C
【题型】主旨大意题
【定位】第⑩段末句
【解析】与末句完全对应。A/B/D 均偏。故选 C。

词汇注释（超纲词）

- **crescent** [n.] 月牙，新月
- **gibbous** [adj.] 凸月的
- **phase** [n.] 月相；阶段
- **orbit** [n./v.] 轨道；绕……运行